Queueing Theory: Part 2

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References

- S.M. Ross, “Introduction to Probability Models”: Chapter 8
- D. Bertsekas and R. Gallager, “Data Networks”: Chapter 3
- R. Wolff: Chapter 5

A Basic Cost Identity

- Consider a queueing system in which each arriving customer pays money according to some rule to the system
- Let λ be the arrival rate of customers
- Then we have the following cost identity:

  1) $\text{average rate at which the system earns} = \lambda \times (\text{average amount an entering customer pays})$

- **Proof**:
  - Let:
    - R_j: amount paid by jth customer, $j \in \{1,2, \dots\}$
    - $R(t)$: total amount earned by system up to time t
    - $\Lambda(t)$: number of arrivals up to time t
  - Then $R(t)$:
    - $\sum_{j=1}^{\Lambda(t)} R_j$
  - So:
    - $\frac{R(t)}{t} = \left(\frac{\Lambda(t)}{t}\right)\left(\frac{\sum_{j=1}^{\Lambda(t)} R_j}{\Lambda(t)}\right)$
  - Result follows by taking limits as $t \to \infty$

Work

- For an arbitrary queueing system, the *work* in the system at time t is defined to be:
  - sum of the remaining service times of all customers in the system at time t
- E.g.:
  - Suppose there are three customers in the system at some time t:
    - The one in service has been there for three of its required five units of service time
    - Both customers in queue have service times of six units
  - Work at time t:
    - $2 + 6 + 6 = 14$
- Let V: average work in the system

A Basic Queuing Identity

- Recall:
  1) average rate at which the system earns $= \lambda \times$ (average amount an entering customer pays)
  - Work in the system at time t is sum of the remaining service times of all customers in the system at time t
- Consider the following cost rule:
  - Each customer pays at a rate of y/unit time when its remaining service time is y, whether it is in queue or in service
- Rate at which system earns at time t:
  - work in the system at time t
- So, by 1):
  2) $V = \lambda E[\text{amount paid by a customer}]$
- Let:
  - S: service time of a given customer
  - D: delay in queue of a given customer
- Then $E[\text{amount paid by a customer}]$:
  - $E\left[SD + \int_0^S (S - x)dx\right]$
  3) $E\left[SD + \frac{S^2}{2}\right]$
- By 2) and 3):
  4) $V = \lambda E[SD] + \frac{\lambda E[S^2]}{2}$
- If a customer's service time is independent of its delay in queue, then:
  5) $V = \lambda E[S]d + \frac{\lambda E[S^2]}{2}$,
  - where d is the average delay in queue

M/G/1 Queue

- Recall:

  1) $V = \lambda E[S]d + \frac{\lambda E[S^2]}{2}$

- Consider an M/G/1 queue
- For an arbitrary customer:

  2) $\text{Customer's delay in queue} =$
     $\text{Work in the system when it arrives}$

- Taking expectations on both sides of 2), we get:

  3) $d = \text{Average work as seen by an arrival}$

- By 3) and PASTA property:

  4) $d = V$

- Using 1) and 4), and solving for d, we get:

  5) $d = \frac{\lambda E[S^2]}{2(1-\lambda E[S])}$

  ❑known as the *Pollaczek-Khintchine formula*

M/G/1 Queue (contd.)

- Recall: Pollaczek-Khintchine formula:

  1) $d = \frac{\lambda E[S^2]}{2(1-\lambda E[S])}$

- Average number of customers in queue:

  2) $Q = \lambda d = \frac{\lambda^2 E[S^2]}{2(1-\lambda E[S])}$, by Little's law

- Average waiting time in system:

  3) $w = d + E[S] = \frac{\lambda E[S^2]}{2(1-\lambda E[S])} + E[S]$

- Average number of customers in system:

  4) $L = \lambda w = \frac{\lambda^2 E[S^2]}{2(1-\lambda E[S])} + \lambda E[S]$, by Little's law

- Condition for d, Q, w, and L to be finite:

  5) $\lambda < \frac{1}{E[S]}$

- For fixed $E[S]$, how do d, Q, w, and L vary with $\mathrm{var}(S)$?
- Recall:

  6) $E[S^2] = \mathrm{var}(S) + (E[S])^2$

- By 1) to 4) and 6), it follows that:
  - ❑ for fixed $E[S]$, d, Q, w, and L increase in $\mathrm{var}(S)$

Example

- Customers arrive to a single server system in accordance with a Poisson process with rate λ
- Each arrival is independently of type $i \in \{1, \dots, n\}$ w.p. α_i, where $\sum_{i=1}^{n} \alpha_i = 1$
- Service time of type i customer has CDF F_i with mean μ_i and variance σ_i^2
- Want to find:
  - Average amount of time a type i customer spends in the system, for $i \in \{1, \dots, n\}$
  - Average number of type i customers in the system, for $i \in \{1, \dots, n\}$
- This model is a special case of the M/G/1 queue
- CDF of service time, S, of a random customer:
  - $G(x) = \sum_{i=1}^{n} \alpha_i F_i(x)$
- $E[S]$:
  - $\sum_{i=1}^{n} \alpha_i \mu_i$
- $E[S^2]$:
  - $\sum_{i=1}^{n} \alpha_i E\{S^2 | \text{type } i\}$
  - $\sum_{i=1}^{n} \alpha_i \left(\mu_i^2 + \sigma_i^2\right)$

Example (contd.)

- Recall:
  - $E[S] = \sum_{i=1}^{n} \alpha_i \mu_i$
  - $E[S^2] = \sum_{i=1}^{n} \alpha_i \left(\mu_i^2 + \sigma_i^2\right)$
- Average delay of type i customer in queue, say d_i:
  - $\text{average work seen by a type } i \text{ customer}$
  - V (by the PASTA principle since type i customers arrive according to a Posson process with rate $\lambda\alpha_i$)
  - $\frac{\lambda E[S^2]}{2(1-\lambda E[S])}$ (by the Pollaczek-Khintchine formula)
  - $\frac{\lambda \sum_{j=1}^{n} \alpha_j \left(\mu_j^2 + \sigma_j^2\right)}{2\left(1-\lambda \sum_{j=1}^{n} \alpha_j \mu_j\right)}$
- Average amount of time a type i customer spends in the system, say w_i:
  - $d_i + \mu_i$
- Average number of type i customers in the system:
  - $\lambda\alpha_i w_i$, by Little's law

M/G/1 Queue (contd.)

- System alternates between idle periods and busy periods
- Want to find expected length of a busy period
- Let:
  - I: length of an idle period
  - B: length of a busy period
  - P_0: long-run proportion of time the system is empty
- Since the process is an alternating renewal process, P_0:

  1) $\frac{E[I]}{E[I]+E[B]}$
- Average number of busy servers in terms of P_0:

  2) $1 - P_0$
- Average number of busy servers in terms of $E[S]$:

  3) $\lambda E[S]$, by Little's law
- By 2) and 3):

  4) $P_0 = 1 - \lambda E[S]$
- $E[I]$:

  5) $\frac{1}{\lambda}$
- Using 1), 4), 5), and solving for $E[B]$, we get:

  6) $E[B] = \frac{E[S]}{1-\lambda E[S]}$

M/G/1 Queue (contd.)

- Recall:

  4) $P_0 = 1 - \lambda E[S]$

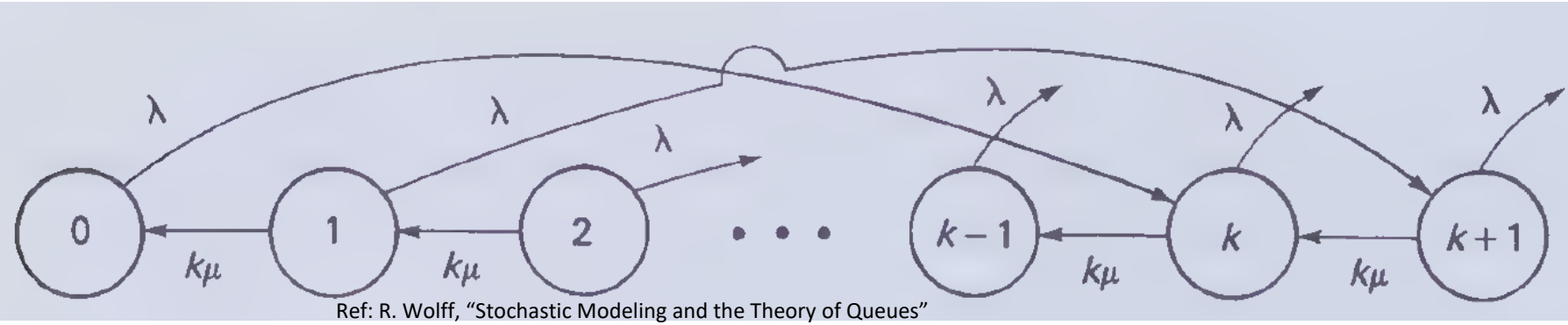
- Let C be the number of customers served in a busy period
- Want to find $E[C]$
- Out of the customers served in a busy period, exactly one finds the system empty when it arrives
- So, by SLLN and PASTA property:
  - $P_0 = \frac{1}{E[C]}$
- Hence, by 4):
  - $E[C] = \frac{1}{1-\lambda E[S]}$

k-Erlang Distribution

- A k-Erlang distribution is a k-fold convolution of an exponential distribution, i.e.:
  - distribution of the sum of k i.i.d. exponential r.v.s
- E.g., a k-Erlang service time, S, can be represented as the sum:
  - $S = \sum_{i=1}^{k} Y_i$,
  - where Y_i, $i \in \{1, \dots, k\}$, are i.i.d. and each is $\exp(k\mu)$
- $E(S)$:
  - $kE(Y_1)$
  - $\frac{1}{\mu}$
- $\mathrm{var}(S)$:
  - $k\mathrm{var}(Y_1)$
  - $\frac{1}{k\mu^2}$
- Distributions of service times and/ or interarrival times of queueing systems are often modeled as k-Erlang since this allows the use of a CTMC model
- If service time distribution is k-Erlang, then it is useful to think of Y_i, $i \in \{1, \dots, k\}$, as service phases that are performed in sequence, e.g.:
  - after a time Y_1, the first phase is completed and $k-1$ phases are remaining
  - if $k-j$ phases are completed, then j phases are remaining, and hence remaining service distribution is j-Erlang

Example: The M/E_k/1 Queue

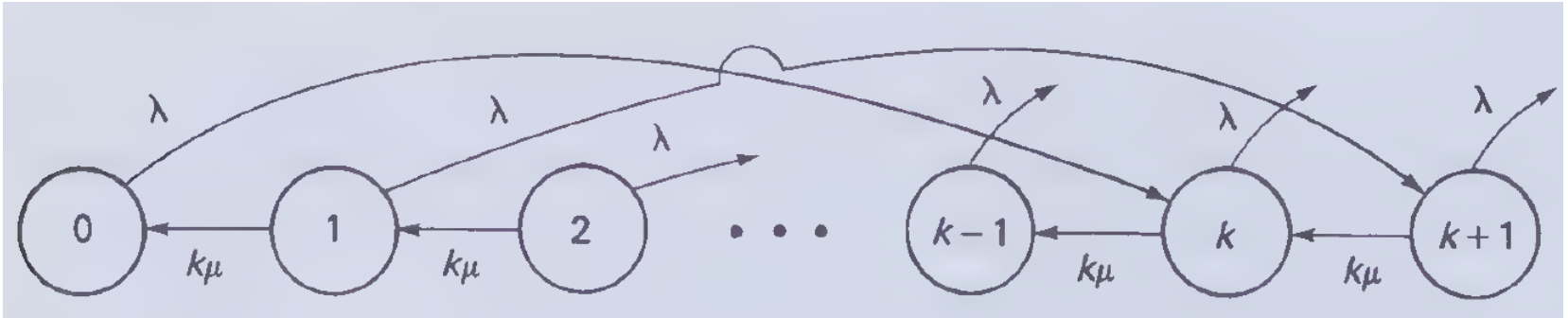
- Recall: a k-Erlang service time, S, can be represented as the sum:
  1) $S = \sum_{i=1}^{k} Y_i$,
  - where Y_i, $i \in \{1, \dots, k\}$, are i.i.d. and each is exp$(k\mu)$
- Suppose:
  - arrival process is Poisson(λ)
  - service times are as in 1)
- A valid state space is:
  - $\mathcal{S} = \{(n, j)\}$,
  - where n is the number of customers in the system and, for $n > 0$, $j \in \{1, \dots, k\}$ is the number of remaining phases associated with the customer in service
  - for $n = 0$, we define $j = 0$
- Number of phases in system:
  - $m = k\max(n-1, 0) + j$
- So, state space can be considered as $\mathcal{S} = \{m : m \in \{0, 1, \dots\}\}$ and transition diagram is as shown in Fig.



Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Example: The M/E_k/1 Queue (contd.)

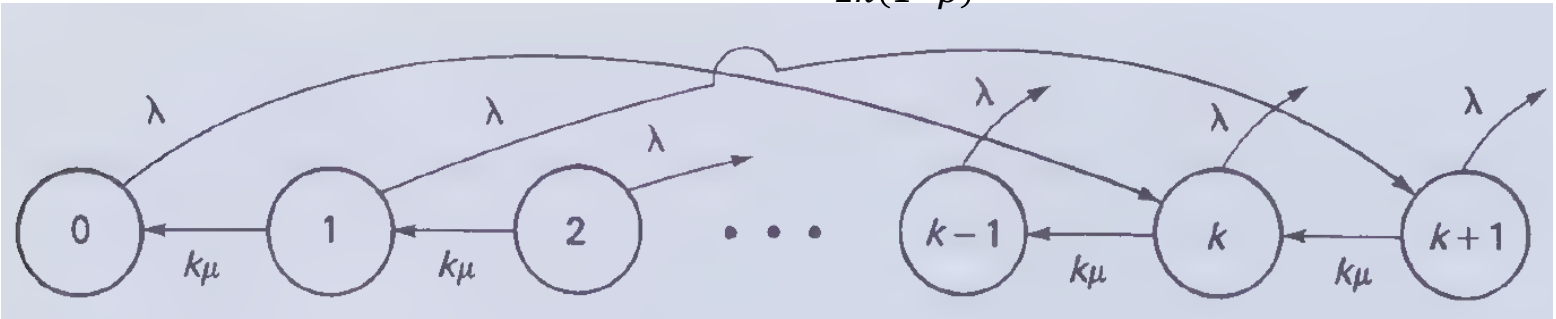
- Let q_m be PMF of number of phases in system
- Balance equations:
  1) $\lambda q_0 = k\mu q_1$
  2) $(\lambda + k\mu)q_m = k\mu q_{m+1}, m \in \{1, \dots, k-1\}$
  3) $(\lambda + k\mu)q_m = k\mu q_{m+1} + \lambda q_{m-k}, m \in \{k, k+1, \dots\}$
- Consider the generating function:
  - ❑ $G(z) = \sum_{m=0}^{\infty} q_m z^m$
- Multiplying 1), 2), and 3) by z, z^{m+1}, and z^{m+1}, respectively, and adding, we get:
  4) $(\lambda + k\mu)zG(z) - k\mu z q_0 = k\mu G(z) - k\mu q_0 + \lambda z^{k+1} G(z)$
- Solving for $G(z)$, we get:
  5) $G(z) = \frac{kq_0(1-z)}{\rho z^{k+1} - (\rho+k)z + k}$,
  - ❑ where $\rho = \frac{\lambda}{\mu}$
- Using $\lim_{z \to 1} G(z) = 1$ and 5), we get:
  6) $q_0 = 1 - \rho$, and
  7) $G(z) = \frac{k(1-\rho)(1-z)}{\rho z^{k+1} - (\rho+k)z + k}$



Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Example: The M/E_k/1 Queue (contd.)

- Recall:
  - *6)* $q_0 = 1 - \rho$, and
  - *7)* $G(z) = \frac{k(1-\rho)(1-z)}{\rho z^{k+1} - (\rho+k)z + k}$
  - ❑ Number of phases in system:
    - *8)* $m = k\max(n-1, 0) + j$
- By 6), system is stable iff:
  - ❑ $\rho < 1$
- By 7), average number of phases, $L_p = \sum_{m=0}^{\infty} m q_m$:
  - ❑ $\lim_{z \to 1} G'(z)$
  - *9)* $\frac{\rho(k+1)}{2(1-\rho)}$
- Writing 8) in terms of r.v.s, we get:
  - ❑ $M = kN_q + J$,
  - ❑ where $L_p = E(M)$ and $Q = E(N_q)$
- Average number of customers:
  - ❑ $L = Q + \rho$
  - *10)* $\frac{(L_p - E(J))}{k} + \rho$
- To find $E(J)$, note that:
  - ❑ every customer eventually passes through every service phase
  - ❑ $P(J = j)$ is the fraction of time there is a customer in phase $k - j + 1$
- By Little's law:
  - ❑ $P(J = j) = \lambda E(\text{time in phase } (k - j + 1)) = \frac{\rho}{k}, j \in \{1, \dots, k\}$
- Hence:
  - 11) $E(J) = \frac{\rho(k+1)}{2}$
- By 9), 10), and 11), we get:
  - ❑ $L = \frac{\rho^2(k+1)}{2k(1-\rho)} + \rho$



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M/G/1 Queue with Batch Arrivals

- Recall the basic queueing identity:

  1) $V = \lambda E[S]d + \frac{\lambda E[S^2]}{2}$

- Suppose arrivals occur in accordance with a Poisson process with rate λ
- Each arrival consists of a batch of a random number of customers:
  - α_j: probability that a batch consists of j customers
  - N: a r.v. representing size of a batch; so, $P(N = j) = \alpha_j$
- Arrival rate of customers:
  - $\lambda E(N)$
- Replacing λ with $\lambda E(N)$ in 1), we get:
  - $V = \lambda E(N)\left[E[S]d + \frac{E[S^2]}{2}\right]$
- For an arbitrary customer:
  - Its delay in queue = work in system when it arrives + its waiting time due to those in its batch
- Taking expectations and using PASTA property, we get:
  - $d = V + E[d_B]$,
  - where d_B is waiting time due to those in its batch

M/G/1 Queue with Batch Arrivals (contd.)

- Recall:

  1) $V = \lambda E(N)\left[E[S]d + \frac{E[S^2]}{2}\right]$

  2) $d = V + E[d_B]$,

  - where d_B is waiting time due to those in its batch
- Recall: during our discussion of batch size biasing, we showed that fraction of customers that belong to batches of size j is given by:
  - $\frac{j\alpha_j}{E(N)}$
- $E[d_B]$:

  3) $\sum_j E(d_B|\text{batch of size } j)\frac{j\alpha_j}{E(N)}$
- $E(d_B|\text{batch of size } j)$:
  - $\sum_{i=1}^{j}(i-1)E(S)\frac{1}{j}$

  4) $\frac{j-1}{2}E(S)$
- Using 3) and 4) and simplifying, we get:

  5) $E[d_B] = \frac{E(S)\left(E(N^2)-E(N)\right)}{2E(N)}$

M/G/1 Queue with Batch Arrivals (contd.)

- Recall:

  1) $V = \lambda E(N)\left[E[S]d + \frac{E[S^2]}{2}\right]$

  2) $d = V + E[d_B]$

  5) $E[d_B] = \frac{E(S)\left(E(N^2) - E(N)\right)}{2E(N)}$

- Substituting 5) into 2) and solving the resulting equation and 1), we get:

  - $d = \frac{\frac{E(S)\left(E(N^2) - E(N)\right)}{2E(N)} + \frac{\lambda E(N)E[S^2]}{2}}{1 - \lambda E(N)E(S)}$

- The queue is stable iff:

  - $\lambda E(N) < \frac{1}{E(S)}$, i.e., arrival rate is less than service rate

- L, Q, and w can be obtained using:
  - $w = d + E(S)$
  - $L = \lambda E(N)w$
  - $Q = \lambda E(N)d$

Priority Queues

- Suppose there are two types of customers, which:
  - arrive according to independent Poisson processes with rates λ_1 and λ_2
  - have service distributions G_1 and G_2
- Type 1 customers are given priority over type 2 customers:
  - service will never begin on a type 2 customer if a type 1 customer is waiting
  - However, if a type 2 customer is being served and a type 1 customer arrives, then the service of the type 2 customer is continued until completion:
    - i.e., there is *no preemption* once service has begun
- Want to find d_i, the average delay in queue of type $i \in \{1,2\}$ customers

Priority Queues (contd.)

- Note that server is always busy whenever there is at least one customer in system
- Total work in the system at any time is exactly the same regardless of which priority rule is used because work:
  - always decreases at a rate of one per unit time when server is busy,
  - always jumps by service time of an arrival
- Hence, work in above system is same as that of an M/G/1 system with:
  - arrival rate $\lambda = \lambda_1 + \lambda_2$,
  - service distribution $G(x) = \frac{\lambda_1}{\lambda} G_1(x) + \frac{\lambda_2}{\lambda} G_2(x)$
- By Pollaczek-Khintchine formula, average work in the priority queuing system is given by V:
  - $\frac{\lambda E[S^2]}{2(1-\lambda E[S])}$
  - $\frac{\lambda_1 E(S_1^2)+\lambda_2 E(S_2^2)}{2(1-\lambda_1 E(S_1)-\lambda_2 E(S_2))}$,
  - where S_i has distribution G_i, for $i \in \{1,2\}$

Priority Queues (contd.)

- Recall:

  1) $V = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1 - \lambda_1 E(S_1) - \lambda_2 E(S_2))}$
- Let V^i be average amount of type $i \in \{1,2\}$ work in system
- Then as in earlier derivation:

  2) $V^i = \lambda_i E[S_i] d_i + \frac{\lambda_i E[S_i^2]}{2}$, for $i \in \{1,2\}$
- Let:

  3) $V_Q^i = \lambda_i E[S_i] d_i$

  4) $V_S^i = \frac{\lambda_i E[S_i^2]}{2}$
- Then V_Q^i (respectively, V_S^i) is average amount of type i work in queue (respectively, service)
- For an arbitrary type 1 customer:
  - Its delay in queue = amount of type 1 work in system when it arrives + amount of type 2 work in service when it arrives
- Taking expectations and using the PASTA property, we get:
  - $d_1 = \lambda_1 E[S_1] d_1 + \frac{\lambda_1 E[S_1^2]}{2} + \frac{\lambda_2 E[S_2^2]}{2}$
- Hence:

  5) $d_1 = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1 - \lambda_1 E(S_1))}$
- Note that $V = V^1 + V^2$; using 1), 2), and 5), and simplifying, we get:

  6) $d_2 = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1 - \lambda_1 E(S_1) - \lambda_2 E(S_2))(1 - \lambda_1 E(S_1))}$

Priority Queues (contd.)

- Recall:

  5) $d_1 = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1-\lambda_1 E(S_1))}$

  6) $d_2 = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1-\lambda_1 E(S_1) - \lambda_2 E(S_2))(1-\lambda_1 E(S_1))}$

- Condition for d_1 to be finite:

  7) $\lambda_1 E(S_1) < 1,$

  - which is independent of type 2 parameters

- Condition for d_2 to be finite:

  8) $\lambda_1 E(S_1) + \lambda_2 E(S_2) < 1$

- Recall: average service time of a customer is:

  9) $\frac{\lambda_1}{\lambda} E(S_1) + \frac{\lambda_2}{\lambda} E(S_2)$

- By 9), it follows that condition in 8) is that the average arrival rate be less than the average service rate

Priority Queues (contd.)

- Recall:

  1) $V^i = \lambda_i E[S_i] d_i + \frac{\lambda_i E[S_i^2]}{2}$, for $i \in \{1,2\}$

  2) $d_1 = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1 - \lambda_1 E(S_1))}$

- Consider the generalization of above system to n types of customers
- Total amount of work in system of customers of types $1, \ldots, j$:
  - is independent of internal priority rule concerning types $1, \ldots, j$
  - only depends on the fact that each of them is given priority over any customers of types $j+1, \ldots, n$
- Hence, $V^1 + \cdots + V^j$ is the same as it would be if:
  - types $1, \ldots, j$ were considered as a single type I priority class, and
  - types $j+1, \ldots, n$ were considered as a single type II priority class

Priority Queues (contd.)

- Recall:

  1) $V^i = \lambda_i E[S_i] d_i + \frac{\lambda_i E[S_i^2]}{2}$, for $i \in \{1,2\}$

  2) $d_1 = \frac{\lambda_1 E(S_1^2) + \lambda_2 E(S_2^2)}{2(1-\lambda_1 E(S_1))}$

- By 1) and 2) applied to above system with type I and II customers:
  - $V^I = \frac{\lambda_I E(S_I^2) + \lambda_I \lambda_{II} E[S_I] E(S_{II}^2)}{2(1-\lambda_I E(S_I))}$,
  - where:
    - $\lambda_I = \lambda_1 + \cdots + \lambda_j$,
    - $\lambda_{II} = \lambda_{j+1} + \cdots + \lambda_n$,
    - $E[S_I] = \sum_{i=1}^{j} \frac{\lambda_i}{\lambda_I} E[S_i]$,
    - $E[S_I^2] = \sum_{i=1}^{j} \frac{\lambda_i}{\lambda_I} E[S_i^2]$,
    - $E[S_{II}^2] = \sum_{i=j+1}^{n} \frac{\lambda_i}{\lambda_{II}} E[S_i^2]$
- Since $V^I = V^1 + \cdots + V^j$, we have an expression for $V^1 + \cdots + V^j$, for each $j \in \{1, \dots, n\}$
- This system of equations can be solved to get $V^1, \dots, V^n$
- We can then use 1) to obtain d_i
- Using this procedure, whose details are left as an exercise, we get:
  - $d_i = \frac{\lambda_1 E(S_1^2) + \dots + \lambda_n E(S_n^2)}{2 \prod_{j=i-1}^{i} \left(1 - \lambda_1 E(S_1) - \cdots - \lambda_j E(S_j)\right)}$, for $i \in \{1, \dots, n\}$